CS 422 Assignment 4: Maximum Concurrent Flow and Network Design

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Code repository: https://github.com/vrajpatel-18/CS422-hw4

1 Part 1: Uniform Traffic Matrix

1.1 Answer

Number the nodes $0, \ldots, n-1$ and let $k = (n-1)/d$. Every node uses the same d *jumps*: node i links to $(i + j) \bmod n$ for each jump j, where the jumps are

$$J = \{\, x \in \{1, \ldots, n-1\} \;:\; x^d \equiv 1 \pmod{n} \,\}.$$

For n prime, J has exactly d elements (this is where $d \mid n-1$ is used). No topology can beat

$$\theta^* \;=\; \frac{k}{2k-1},$$

and our topology reaches it whenever every node is within 2 hops of every other. For example, $n = 13$, $d = 4$ gives $J = \{1, 5, 8, 12\}$, every node is within 2 hops (checked in Section 1.5), and $\theta = \theta^* = 3/5$.

Scope. The upper bound holds for every n and every topology. Finding the optimal topology for every n and d is closely tied to the degree–diameter problem (how many nodes a degree-d graph can fit within a given number of hops), which is a long-standing open problem. So the answer here is the bound, plus a construction that meets it exactly for prime n whenever it has every node within 2 hops (Section 1.6).

1.2 Setup

Each node has d outgoing and d incoming links of capacity 1, so the network has nd units of capacity in total. Each node sends $\frac{d}{n-1}$ to each of the other $n-1$ nodes. The *concurrent flow* θ is the largest factor for which θT fits on the links. Notation:

- $\text{hops}(s, t)$: the fewest links needed to get from s to t.
- A: the average of $\text{hops}(s, t)$ over all $n(n-1)$ ordered pairs.
- $k = (n-1)/d$: the number of destinations per outgoing link (an integer by assumption).

1.3 Upper bounds that hold for every topology

Bound 1. *For every topology,* $\theta \le 1/A$.

Proof. Each unit of traffic from s to t crosses at least $\text{hops}(s,t)$ links and uses one unit of capacity on each. So the scaled traffic uses at least

$$\theta \cdot \frac{d}{n-1} \cdot \sum_{s \neq t} \text{hops}(s,t) \;=\; \theta \cdot nd \cdot A$$

units of capacity. Only nd units exist, so $\theta\, nd\, A \le nd$, i.e. $\theta \le 1/A$. □

Bound 1 is met with equality only if (i) all traffic follows shortest routes and (ii) every link is completely full. So a good topology needs **few hops** and **evenly spread load**.

Bound 2. *For every topology,* $A \ge \frac{2k-1}{k}$, *so* $\theta \le \frac{k}{2k-1}$.

Proof. A node has only d outgoing links, so at most d of its $n-1$ destinations, a fraction $1/k$, are 1 hop away. All others are at least 2 hops away. So $A \ge \frac{1}{k} \cdot 1 + \frac{k-1}{k} \cdot 2 = \frac{2k-1}{k}$, and Bound 1 gives $\theta \le \frac{k}{2k-1}$. □

$k = (n-1)/d$	1	2	3	4	large
best possible θ^*	1	2/3	3/5	4/7	$\to 1/2$

The case $k = 1$ ($d = n-1$) is the complete graph: every pair has a direct link and $\theta^* = 1$.

1.4 The construction

Algorithm 1 BUILDTOPOLOGY(n, d), for n prime and $d \mid n-1$

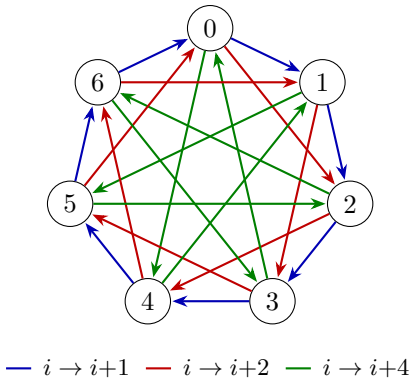
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J ← { x ∈ {1, ..., n−1} : x^d mod n = 1 }        ▷ exactly d jumps
for each node i = 0, ..., n − 1 do
    for each jump j ∈ J do
        add link i → (i + j) mod n                 ▷ capacity 1
Routing: split the traffic of each pair evenly over all its shortest routes.
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Why it is a valid topology. Node i has d outgoing links (one per jump) and d incoming links (from $i - j$ for each jump j). Jumps are distinct and nonzero, so there are no self-loops and no duplicate links. For n prime, $x^d \equiv 1$ has exactly d solutions when $d \mid n-1$ (the nonzero numbers mod a prime form a cyclic group of size $n-1$), so $|J| = d$.

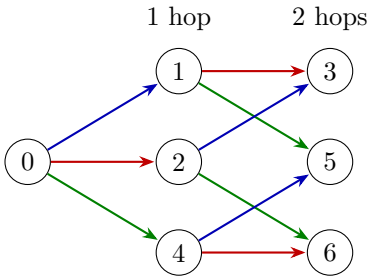
The key property. The jumps are *closed under multiplication*: if $x^d \equiv 1$ and $y^d \equiv 1$ then $(xy)^d \equiv 1$. Likewise, the inverse of a jump is a jump. For $n = 7$, $d = 3$: $J = \{1, 2, 4\}$, and $2 \cdot 2 = 4$, $2 \cdot 4 = 8 \equiv 1$, $4 \cdot 4 = 16 \equiv 2$. Finding J by hand is easy: $1^3 = 1$, $2^3 = 8 \equiv 1$, $4^3 = 64 \equiv 1 \pmod 7$, but $3^3 = 27 \equiv 6$, so $3 \notin J$.

1.5 Why it maximizes the concurrent flow

This section shows the construction meets both bounds with equality. Claim 1 handles Bound 1 (the load is perfectly even), and Claim 2 handles Bound 2 (the hop count is as small as possible).



(a) the topology: jumps $\{1, 2, 4\}$



(b) shortest routes from node 0

Figure 1: $n = 7$, $d = 3$, $J = \{1, 2, 4\}$. (a) The topology; colours mark the jump of each link. (b) From node 0, the 3 jumps are 1 hop away and the other 3 nodes are 2 hops away, each reached by exactly two shortest routes ($3 = 1 + 2 = 2 + 1$, $5 = 1 + 4 = 4 + 1$, $6 = 2 + 4 = 4 + 2$). So $A = 1.5$ and $\theta = 2/3$, which matches Bound 2.

What a relabeling is. A relabeling renames every node, giving node x the new name $\sigma(x)$. We use relabelings of one form:

$$\sigma(x) = a \cdot x + b \pmod{n}, \qquad a \in J,\ b \text{ any number in } \{0, \ldots, n-1\}.$$

In words: multiply every node's number by a jump a, then shift every number by b. Each node gets a different new name, because n is prime, so multiplying by $a \neq 0$ can be undone. We say a relabeling *maps the topology onto itself* if, after renaming, the set of links is exactly the same as before. When that happens, nothing about the network can tell two links apart: if one link had some special feature, the relabeling would carry that feature to another link.

A worked example ($n = 7$, $J = \{1, 2, 4\}$). Take $a = 4$, $b = 2$, so $\sigma(x) = 4x + 2 \bmod 7$:

old name x	0	1	2	3	4	5	6
new name $\sigma(x)$	2	6	3	0	4	1	5

The link $0 \to 1$ (jump 1) becomes $2 \to 6$, which is a link (jump 4). The link $0 \to 4$ (jump 4) becomes $2 \to 4$, also a link (jump 2). Every link maps to a link this way. This particular σ also sends the link $3 \to 5$ onto the link $0 \to 1$: $\sigma(3) = 0$ and $\sigma(5) = 1$.

Claim (1: every link carries the same load, so $\theta = 1/A$)**.**

Proof. We check three facts about the relabelings $\sigma(x) = ax + b$.

- *Each one maps the topology onto itself.* The link $x \to x + j$ is renamed to $\sigma(x) \to \sigma(x) + aj$. Since a and j are both jumps, aj is a jump (the key property), so the renamed link is a real link.
- *Any link can be moved onto any other.* To move the link $i \to i + j$ onto $0 \to 1$, use $a = j^{-1}$ (a jump, since inverses of jumps are jumps) and $b = -j^{-1}i$. Then $\sigma(i) = 0$ and $\sigma(i + j) = j^{-1}j = 1$. In the example above, $j = 2$, $j^{-1} = 4$ and $i = 3$ give exactly $\sigma(x) = 4x + 2$. Any link can be moved to $0 \to 1$ and back out again, so any link can be moved onto any other.

- *Nothing else changes either.* The traffic is the same amount between every pair, and the routing rule (split evenly over shortest routes) only looks at hop counts. Renaming nodes changes neither.

Now suppose link e carries some load. Pick a relabeling that moves e onto another link e'. After renaming, the network, the traffic and the routing are all exactly as before, so e' must carry the same load that e did. Hence all nd links carry the same load. The total load is $\theta \cdot nd \cdot A$ (from the proof of Bound 1), so each link carries θA. Setting $\theta = 1/A$ fills every link exactly to capacity. That routing is feasible, and Bound 1 says nothing larger is, so $\theta = 1/A$. □

Claim (2: within 2 hops, A meets Bound 2)**.**

Proof. The 1-hop destinations of a node are exactly its d jumps, the most Bound 2 allows. If every other node is a sum of two jumps, it is exactly 2 hops away, so $A = \frac{2k-1}{k}$ and $\theta = \frac{k}{2k-1}$, the best possible. Every node sees the same picture as node 0 (rename with $a = 1$, which just shifts every number by b), so checking node 0 suffices. □

Putting the two claims together: whenever every node is within 2 hops, our topology achieves $\theta^* = \frac{k}{2k-1}$, and no topology can do better.

Checking the 2-hop condition by hand. By Claim 2, it is enough to check that from node 0, every node that is not a jump is a sum of two jumps (mod n).

- $n = 7$, $d = 3$, $J = \{1, 2, 4\}$: the sums of two jumps are $\{1, 2, 3, 4, 5, 6\}$ (mod 7), which cover all of $\{3, 5, 6\}$ (Figure 1). So $\theta = 2/3 = \theta^*$.
- $n = 13$, $d = 4$, $J = \{1, 5, 8, 12\}$. Ignoring 0, the sums of two jumps mod 13 are $\{2, 3, 4, 6, 7, 9, 10, 11\}$. Together with J this is all 12 other nodes. So $\theta = 3/5 = \theta^*$.

Why both properties matter ($n = 13$, $d = 4$).

- Jumps $\{1, 2, 3, 4\}$: from node 0, four nodes each are at 1, 2 and 3 hops, so $A = 2$ and $\theta \leq 1/2$ by Bound 1. Too many hops.
- Jumps $\{1, 2, 6, 9\}$: also within 2 hops (the sums of two jumps give $\{2, 3, 4, 5, 7, 8, 10, 11, 12\}$), with the same $A = 5/3$ as ours. But $\{1, 2, 6, 9\}$ is not closed under multiplication ($2 \cdot 2 = 4 \notin J$), so the relabelings of Claim 1 do not map this graph onto itself. Nothing then forces every link to carry the same load, and Bound 1 is only met if every link is exactly full. Few hops alone are not enough.
- Our jumps $\{1, 5, 8, 12\}$: within 2 hops, and every link carries the same load (Claim 1), so $\theta = 3/5 = 0.600$, the best possible.

1.6 Limitations

- n **must be prime** for this construction. Only then is $|J| = d$ guaranteed, and only then does every relabeling in Claim 1 give each node a different name. Bounds 1 and 2 hold for every n.

- **Very sparse networks.** A node reaches at most d nodes in 1 hop and d^2 in 2 hops. If $d+d^2 < n-1$, not everyone can be within 2 hops, so no topology reaches Bound 2. (Counting d^3 nodes at 3 hops, and so on, gives a tighter bound.) Even when $d + d^2 = n - 1$ exactly, every 2-hop route would have to land on a different node, and it is known that no such graph exists for $d \geq 2$. For example, $n = 13$, $d = 3$ has $d + d^2 = 12 = n - 1$, so no topology reaches Bound 2 there.
- **The jumps do not always give 2 hops.** Even when the counting above allows 2 hops, our jumps can need 3. For $n = 17$, $d = 4$ ($d + d^2 = 20 \geq 16$), the jumps are $\{1, 4, 13, 16\}$, and the sums of two jumps (mod 17, ignoring 0) are $\{2, 3, 5, 8, 9, 12, 14, 15\}$. Together with J that is only 12 of the 16 other nodes: $6, 7, 10$ and 11 are 3 hops away. Then $A = (4{\cdot}1+8{\cdot}2+4{\cdot}3)/16 = 2$, so our $\theta = 1/2$ falls short of $k/(2k-1) = 4/7$.
- In both of these cases the load is still perfectly even ($\theta = 1/A$ exactly, by Claim 1), so the only loss comes from the extra hops.

2 Part 2: Arbitrary Traffic Matrix